A058 — Trefoil GUI-Slider Matched-Budget Response Falsifier

SST-Workbench

Version v0.2.10 — SST Falsifier Framework v1.0.8

Machine-readable science contract

Research question How do eight anonymous KnotPlot GUI controls change the same trefoil after matched finite time horizons of 512, 2048 and selectively 32768 go iterations compared with their sector-default references?

Objective Quantify finite-time shape/scalar/parametrization response, reproducible across checkpoints; qualify compatibility and execution failures; do not infer convergence at a fixed budget.

H_0 H0: no registered slider perturbation changes matched-iteration geometry beyond numerical noise under the declared baseline/seed.

H_1 H1: at least one registered slider perturbation yields a measurable shape/scalar/parametrization deviation at a matched checkpoint.

Registered equations

E1: Quarter-octave ladder

$$m_k = 2^{k/4}, \qquad k = -16, \ldots, 16 \tag{1}$$

Dimensional check: Exponent and multiplier are dimensionless.

E2: Log response coordinate

$$x = \log_2 m \tag{2}$$

Dimensional check: Logarithm argument and result are dimensionless.

E3: Normalized shape response

$$D_{\text{shape}}(X, Y) = \min_{R,\delta,\sigma} \left[M^{-1} \sum_i \|\hat{X}_i - R\hat{Y}_{\sigma i+\delta}\|^2 \right]^{1/2} \tag{3}$$

Dimensional check: Normalized coordinate differences are dimensionless.

E4: Matched-iteration observation horizons

$$N \in \{32, 128, 512, 2048, 8192, 32768\} \quad \text{(tier-dependent)} \tag{4}$$

Dimensional check: All counts are dimensionless numbers of solver iterations.

E5: Bead parametrization diagnostic

$$C_{\text{edge}} = \sigma(\Delta s_i)/\langle \Delta s_i \rangle \tag{5}$$

Dimensional check: Ratio of equal-length quantities is dimensionless.

E6: Finite-time matched response instead of convergence-time ratio

$$D_p(m; N) = D_{\text{shape}}\big(X_{p=mp_0}(N), X_{p=p_0}(N)\big) \tag{6}$$

Dimensional check: Both inputs are at matched dimensionless iteration N; normalized shape distance is dimensionless.

E7: Campaign completeness

$$C = N_{\text{valid}}/N_{\text{registered}} \tag{7}$$

Dimensional check: Count ratio is dimensionless.

E8: Sector activation comparison independent of strength sweep

$$D_{\text{act}} = D_{\text{shape}}(X_{R2,\,\text{off}}, X_{R2,\,\text{on}}) \tag{8}$$

Dimensional check: Normalized shape comparison is dimensionless.

E9: Reproducible one-time geometric seed-preparation target

$$d_{\text{target}} = 1.05\, d_{\text{close},0}, \quad d_{\text{close},0} > 0 \tag{9}$$

Dimensional check: Both target minimum separation and the close threshold carry KnotPlot geometry length units; 1.05 is dimensionless.

E10: Blinded high-budget selection score

$$S = 0.7 D_{\text{shape}} + 0.3 \Delta_{\text{scalar}}, \qquad |\mathcal{C}_{32768}| \le 31 \tag{10}$$

Dimensional check: Both response terms and score are dimensionless.

Registered code-to-mathematics steps

1. **S1** [G1; equations E7, E9]: Runtime-probe eight anonymous slider parameters in six enabled sector contexts; verify assignment-form setters and sector switches; for all controls verify numeric getter agreement to six-decimal display precision, including declared getter aliases, and verify sector switch state; verify the Damped + collision-fast regime; and export exactly one safe trefoil during installation. Prepare exactly one shared ASCII safe seed with a one-time geometry-scale normalization at 1.05 times the measured close default; reload it and check safety before and after go 1, while keeping all dynamics/default parameters unchanged.
2. **S2** [G2; equations E1, E2]: Generate six enabled-sector default references, one additional matched R2-OFF activation control, and eight default-relative one-slider-at-a-time sweeps on the quarter-octave ladder with zero endpoints.

3. **S3** [G2; equations E4]: Observe all 271 conditions at go 32,128,512 in BASIC and 32,128,512,2048 in extended tier; no adaptive convergence claim; store separate run identities.

4. **S4** [G3; equations E3, E5, E6]: Compute matched-iteration shape and scalar responses relative to each sector reference; distinguish missing runtime lanes and numerical failures.

5. **S5** [G5; equations E5]: Certify coordinate-derived scalar metrics with C++17/pybind11 FP64 against Python FP64.

6. **S6** [G8; equations E6, E10]: Preselect at most 31 conditions from prior matched-iteration evidence with deterministic selection preserving references and up to three responses per axis; follow to go 32768 with interim checkpoints.

7. **S7** [G9; equations E7]: After commitment-verified reveal only, attach GUI slider and sector names to the frozen anonymous response curves, including the registered Damped-regime caveat for the velocity-dependent sector.

8. **S8** [G3; equations E8]: Execute one separately registered R2 switch-OFF control with the identical S3/S4 numerical default setters as the enabled R2 reference, and compare final shapes without pooling this binary contrast into any slider response fit.

1 Identification

A058 v0.2.10 is a single-trefoil, eight-slider, default-relative KnotPlot response experiment with matched iteration-budget observations with staged horizons instead of claiming convergence. It is pinned to SST Falsifier Framework v1.0.8.

2 Question

Which anonymous GUI slider parameters alter the finite-time embedded geometry and bead parametrization at identical observation counts relative to their sector-context runtime default values?

3 Hypotheses

The frozen science contract registers the global no-effect null and measurable-response alternative.

4 Assumptions

The actual executable parameter dumps define numerical defaults. Sliders may be conditional on an enabled GUI sector and therefore need not appear in the bare reset-all dump. The user-requested dynamics is damped with collision avoidance; the original catalogue embedding itself is NOT an admissible default-safe seed. A once-only geometric safe-seed preparation uses `fitto mindist` at 1.05 times the measured reset-default close threshold. This normalizes the absolute embedding scale but does not modify any dynamics parameter. All conditions reuse the same saved seed without per-condition fitting, refining, or setting close/max-dr/stusplit/time increment.

5 Symbols

Primary quantities are multiplier m, $x = \log_2 m$, normalized shape distance D_{shape}, edge-spacing coefficient of variation, and registered iteration checkpoint N.

6 Equations

Registered equations and dimensional checks are generated from the frozen science contract.

7 Algorithm

Install calibration verifies eight setters in six enabled sector contexts. It normalizes the raw catalogue trefoil once to a registered safe geometry scale, exports exactly one ASCII trefoil, reloads that exported geometry, and checks safety before and after a single dynamics step. Every scientific condition then loads precisely that same immutable seed. All eight axes now require runtime numeric getter evidence using the six-decimal display precision of the KnotPlot parameter dump, including the separately resolved S3 getter alias. The registered reset-default S3=1.40 and R2 reset OFF are preserved (enabled reference ON). This certifies rounded displayed values, not the exact internal FP64 parameter state. The production campaign contains six sector-on references, one separate sector-off R2 activation control, and eight one-slider-at-a-time sweeps. Each scientific condition is advanced to preregistered common observation checkpoints using incremental go commands: BASIC at 32, 128, 512; extended tier additionally at 2048; heavy tier additionally at 8192 and 32768. The heavy tier is restricted to at most 31 automatically preselected conditions from valid earlier results, including independently hash-bound qualification overlays of legacy runs if explicitly imported. Each budget is a finite-time observation horizon, not an attractor or convergence certification.

8 Sources

One runtime-identified KnotPlot executable and one immutable, preconditioned 300-bead trefoil ASCII seed (SHA-256 committed in the acquisition manifest) define the simulation source family. Conditions are perturbations, not independent replicates.

9 Gates

G0–G9 retain Framework-v1.0.8 lane semantics. Numerical failures are recorded and the campaign continues where possible so diagnostic structure is retained.

10 Numerics

Positive multipliers are $2^{k/4}$, $k = -16, \ldots, 16$, with a separate zero endpoint. BASIC executes 512, 2048 executes 2048, and the selected heavy lane executes 32768 matched go iterations; checkpoints are at preregistered counts, not a claim of stability.

11 Backends

Python/NumPy FP64 is the reference analysis. C++17/pybind11 FP64 certifies registered scalar coordinate metrics.

12 Blindness

Public analysis uses anonymous slider IDs S0–S7 and reference IDs R0–R5. GUI names and command mappings reside in the private reveal domain. One additional R2-OFF activation contrast is separately registered and is not pooled with the multiplier responses.

13 Statistics

Quadratic fits in $x = \log_2 m$ are descriptive finite-time fits only. Exact-zero endpoints are excluded from log fits. The primary response measures are final shape, scale-free scalar geometry, bead-spacing variation, and matched-iteration horizon. The R2 ON/OFF contrast at identical numeric slider defaults is analyzed separately.

14 Results

Automatic run provenance

Field	Recorded value
Run ID	20261013T214624700122Z-2461a425
Run phase	BLIND
Protocol bundle SHA-256	12d27f37b175ad3efb61ca70f299f1230ce0ec54b572432bbb66c52836d6715f
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	01db91c00625fa2fb7378fb69f5429c49c584b072ca4e8cb62132c8a2a742920

Gate ledger

Gate	Status	Question
G0	PASS	Are Framework-v1.0.8 and A058-v0.2.10 frozen protocol bytes intact?
G1	PASS	Are user-confirmed reset defaults, exactly one safely preconditioned shared trefo
G2	PASS	Is the complete registered multiplier design valid, and are matched-budget chec
G3	PASS	Do anonymous slider perturbations change geometry at matched time horizons
G4	UNRESOLVED	Is there an independent held-out physical confirmation source?
G5	PASS	Does strict native CPU FP64 certification reproduce Python coordinate metrics
G6	NOT_APPLICABLE	If enabled by profile, does GPU screening satisfy its registered parity contract?
G7	NOT_APPLICABLE	If enabled, is there independent cross-source/multilibrary replication?
G8	PASS	Are all selected observations runtime-verified and collision-safe at their respecti
G9	DEFERRED	After blind gates, do reveal commitments verify without changing frozen blind s

Backend manifest


```
{
  "knotplot_campaign": "single trefoil; 8 anonymous GUI sliders; 6 sector references; conditi
  "native_cpu": {
    "authority": "CERTIFICATION",
    "expected_backend": "C++17/pybind11 FP64",
    "silent_fallback": false
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2"
}
```

Source manifest


```
{
  "manifest_sha256": "ba6841eb5e62fcdbd554a95c001d13ea80a7f549f64781b8f86941cd32f974ed",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A058_HISTORICAL_TREFOIL_SCRIPT",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "KnotPlot_script_family",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A058_KnotPl
      "provenance_family": "historical_KnotPlot_export",
      "registry_hint": {
        "catalog_id": "A058_HISTORICAL_TREFOIL_SCRIPT",
        "hits": []
      },
      "role": "discovery",
      "sha256": "95a4338c254af276e95ba102eed7f9f54a6781be47793c50356cab2623d74517",
      "source_id": "S0",
      "version": "historical"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation

A slider may affect parametrization or convergence time without selecting a different final embedded curve. Conversely, persistent final-shape changes identify a geometry-selecting numerical control in this KnotPlot model. One anonymous lane is explicitly interpreted as a compatibility test because the KnotPlot GUI advises a different damping regime for that force than the user-requested baseline. The findings are conditional on the explicitly normalized common seed scale; changing that seed scale could change forces and the selected attractor. None of these outcomes establishes physical vortex stability by itself.

16 Reproducibility

Canonical instance: `01_research/A_falsifiers/A058_KnotPlot_Parameter_Sensitivity_Relaxation_Robustness/A058-v0.2.10`. Use `run_all.cmdBASIC/2048/32768` for install calibration, campaign generation, freeze, matched-budget acquisition and full Framework analysis.

References

[1] R. G. Scharein, “KnotPlot Manual,” Hypnagogic Software.